

Queues

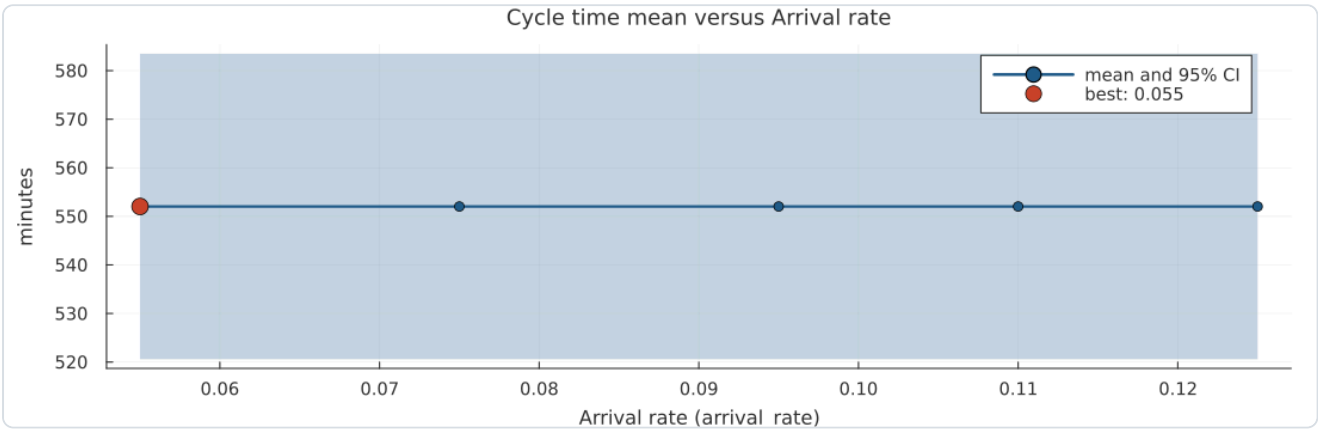
Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 | Generated: | Engine: DiscreteSim | Sections: 8

Queues and capacity

The first thing to check in a queueing model is that it agrees with Erlang, and the second is what happens when the load moves.

Widening the bottleneck



Sweep

Arrival rate

Value	Throughput mean	Wait mean	Utilisation mean	Cycle time mean
0.055	0.133	—	0.865	—
0.075	0.133	—	0.865	—
0.095	0.133	—	0.865	—
0.11	0.133	—	0.865	—
0.125	0.133	—	0.865	—

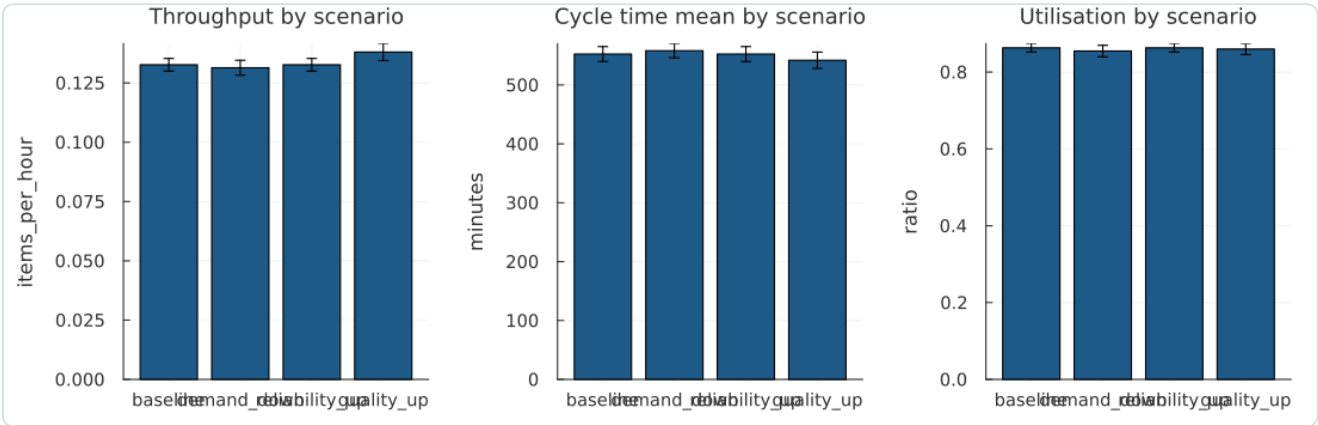
The best arrival_rate by cycle_time_mean is 0.055 (mean 552 ± 31.5).

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.133	0.131	-0.00126	0.00163	-0.953	indistinguishable
demand_down	cycle_time_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	wait_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	utilisation	0.863	0.855	-0.00824	0.00893	-0.955	indistinguishable

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	completed	477.5	473	-4.5	5.844	-0.942	indistinguishable
reliability_up	throughput	0.133	0.133	0	0	0	indistinguishable
reliability_up	cycle_time_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	wait_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	utilisation	0.863	0.863	0	0	0	indistinguishable
reliability_up	completed	477.5	477.5	0	0	0	indistinguishable
quality_up	throughput	0.133	0.138	0.00535	0.00154	4.032	better
quality_up	cycle_time_mean	552.5	541.9	-10.6	5.537	-1.926	better
quality_up	wait_mean	552.5	541.9	-10.6	5.537	-1.926	better
quality_up	utilisation	0.863	0.86	-0.00318	0.00578	-0.369	indistinguishable
quality_up	completed	477.5	496.8	19.2	5.523	4.031	worse



Scenarios

Best by cycle_time_mean: **quality_up** (compared with baseline, common random numbers on every replication).